深圳技术大学
SHENZHEN TECHNOLOGY UNIVERSITY

Recommendation Letter - Academic Advisor

Date: 26 September 2026

Dear Members of the Admissions Committee,

I am writing to support Yongyi Lai's application to the MSc in Artificial Intelligence and Robotics at The Chinese University of Hong Kong, Shenzhen.

I am Associate Dean in the School of Artificial Intelligence at Shenzhen Technology University, where Yongyi is a final-year student in Computer Science and Technology. I have known her since 2023, first as the instructor of University Computer Science A and, since then, as the supervisor of two robotics projects in my laboratory. I have also followed her competition and independent work closely, and this letter draws on all three.

Yongyi first came to my attention as a strong student who deliberately extended a Computer Science background into robotics. Her official cohort standing is 6th among 156 students, approximately top 3.85%. For knowledge, I rank her in the top 5% among the 156 students in her 2023 Computer Science cohort. For suitability for the programme, I rank her in the top 10% among the same 156 students, based on her technical preparation and the robotics project work I have observed.

The first project, a wheel-legged robot with variable leg length, tested this preparation under changing mechanical geometry: the balance controller had to remain effective as the robot's leg configuration changed. Yongyi constructed the body-pitch and motion states from IMU, wheel-speed and joint feedback, implemented an LQR controller with gains scheduled by leg length, and added a virtual model controller that derived virtual leg length and posture from the closed-chain leg kinematics and mapped support force and pitch torque to the joints through the Jacobian. The controller was integrated and tuned on the physical robot, running on an STM32H7/FreeRTOS platform with FDCAN. The robot kept its body balanced and tracked velocity commands while its legs extended and retracted. When balance failed, Yongyi traced the causes to state inconsistency, torque limits and ground-contact transitions, then corrected them systematically.

The second project expanded this work from locomotion to navigation. On a car-like mobile robot, Yongyi was responsible for the autonomy chain: EKF/AMCL localisation fusing IMU, wheel encoders and 2D LiDAR; Hybrid-A* global planning under the vehicle's kinematic constraints; TEB local trajectory optimisation; and an MPC tracking controller on a car-like kinematic model. She addressed timing and coordinate-frame inconsistencies that caused pose jumps. The system ran on the physical vehicle and achieved autonomous navigation from localisation through obstacle-aware planning to trajectory tracking. It is unusual at undergraduate level for a student to take responsibility for work spanning localisation, planning and control and carry it through to a functioning vehicle.

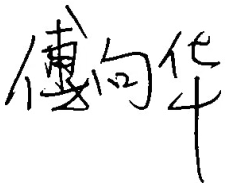
I have also seen the same capabilities in work I did not supervise directly but followed closely. In the university RoboMaster team, Yongyi was responsible for embedded control of the hero robot, implementing gimbal, launcher and chassis control loops on STM32F407/FreeRTOS over CAN. The robot competed in the 2025 national final, where the team won third prize, and the team also took second prize in the 3V3 league. Yongyi also designed and built an OpenHarmony smart gardening system independently: it runs YOLOv5 inference on an RK3588 edge device, makes decisions through a LangGraph agent chain and controls irrigation on a Hi3861 device, completing a perception-to-actuation loop.

Taken together, these experiences show a student who can move from control theory to embedded implementation and back, who finishes systems rather than components, and who treats documentation and hardware safety as part of the design. In both laboratory projects Yongyi derived a model for the particular robot and tuned the controller on the machine. A taught master's is the right next step to deepen her foundations in control, planning and learning so that she can design methods more systematically. I believe she will benefit substantially from your programme.

I recommend Yongyi Lai for admission and would be glad to provide any further information.

Yours sincerely,

Referee signature:



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